

1. The main issue of deadlock detection is _____.
how often to detect deadlock
2. The wait-for graph is a _____.
a detection method less general than the detect_deadlock function (similar to safe)
3. A good time to detect a deadlock would be when _____.
a process just got a resource request denied
4. The deadlock detection function must be called _____.
None of these answers
5. The main issue of deadlock recovery is
how to rollback a terminated process
how to select a process to terminate
how to roll back a process stripped from its resources
6. This system _____ because _____ process(es) can complete.

i	Allocation			Request		
	A	B	C	A	B	C
0	3	0	2	1	3	1
1	3	0	2	7	1	1
2	2	1	1	1	2	2
3	0	0	2	5	4	2
4	0	3	0	8	3	4

Available		
A	B	C
3	4	1

$Work = (3 \ 4 \ 1)$
 $\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ \times & \times & \times & \times & \times \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \end{matrix}$
 ① $(3 \ 4 \ 1) + (3 \ 0 \ 2) = (6 \ 4 \ 3)$
 ② $(6 \ 4 \ 3) + (2 \ 1 \ 1) = (8 \ 5 \ 4)$
 ③ $(8 \ 5 \ 4) + (0 \ 0 \ 2) = (8 \ 5 \ 6)$
 ④ $(8 \ 5 \ 6) + (0 \ 3 \ 0) = (8 \ 8 \ 6)$
 ⑤ $(8 \ 8 \ 6) + (3 \ 0 \ 2) = (11 \ 8 \ 8)$
 is not in deadlock
 all

is not in deadlock
all

7. This system _____ because _____ process(es) can complete.

i	Allocation			Request		
	A	B	C	A	B	C
0	0	3	0	8	3	4
1	3	0	0	1	3	1
2	3	0	2	7	1	1
3	2	1	1	1	2	2
4	0	0	2	5	4	2

Available		
A	B	C
1	3	1

$Work = (1 \ 3 \ 1)$
 $\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ \checkmark & \times & \times & \times & \times \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \end{matrix}$
 ① $(1 \ 3 \ 1) + (3 \ 0 \ 0) = (4 \ 3 \ 1)$
 is in deadlock
 at most 1

8. This system _____ because _____ process(es) can complete.

i	Allocation			Request		
	A	B	C	A	B	C
0	3	0	2	7	1	1
1	2	1	1	1	2	2
2	0	0	2	5	4	2
3	0	3	0	8	3	4
4	1	0	2	1	3	1

Available		
A	B	C
1	2	2

is in deadlock
at most 2

Decide something is in deadlock
 Work = available (122)
 0 1 2 3 4
 F ~~RT~~ F F ~~T~~ ← Finished?
 request ≤ work
 ① (122) + (211) = (333) = work
 ④ (333) + (102) = (435) = work
 No more so deadlock
 2 process finishes

9. Answer all these questions using the Baker's Algorithm. The objective of this exercise is to find out whether a request from Process 4 can be granted with the current state (see below). Process 4 requests the resources as described with this vector (1 3 2). Will the system grant this request?

i	Allocation			Need		
	A	B	C	A	B	C
0	3	0	2	7	1	1
1	2	1	1	1	2	2
2	0	0	2	5	4	2
3	0	3	0	8	3	4
4	3	0	2	1	3	1

Available

A B C
3 4 1

False

10. Answer all these questions using the Baker's Algorithm. The objective of this exercise is to find out whether a request from Process 2 can be granted with the current state (see below). Process 2 requests the resources as described with this vector (4 4 1). Will the system grant this request?

<i>i</i>	Allocation				Need		
	A	B	C		A	B	C
0	3	0	2		7	1	1
1	2	1	1		1	2	2
2	0	0	2		5	4	2
3	0	3	0		8	3	4
4	3	0	2		1	3	1

Available

A	B	C
3	4	1

False